

AWS IAM ACTIVITIES

ACTIVITY 1 – IAM Users & Authentication

Objective

Understand IAM users and authentication mechanisms.

Tasks

1. Create an IAM user named: intern-user1
2. Enable AWS Management Console access
3. Set a custom password
4. Log in using the IAM user and verify:

- ☐ You cannot access S3
- ☐ You cannot access EC2

Expected Outcome

- User exists
- Login works
- Everything shows Access Denied

User named Intern-user1 has been created as an IAM user. Initially EC2 start and stop instances permissions were given under policy where the resource ARN is chatapp-backend. But IAM user is unable to view/start/stop any instance. So, we need to define DescribeAllInstances where the resource is given as * to be able to view all the resource. However, the start / stop instance is only given to chatapp-backend

✔ Policy Intern-user1-policy created.

Permissions

Entities attached

Tags

Policy versions (1)

Last Accessed

Permissions defined in this policy [Info](#)

Copy

Edit

Summary

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Sid": "VisualEditor0",  
6       "Effect": "Allow",  
7       "Action": [  
8         "ec2:StartInstances",  
9         "ec2:StopInstances"  
10      ],  
11      "Resource": "arn:aws:ec2:ap-south-1:053020126602:instance/i-0c0d9260e000ebc44"  
12    }  
13  ]  
14 }
```

[Alt+S]



Asia Pacific (Mumbai)

Account ID: 0530-2012-6602

intern-user1

Instances [Info](#)



Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >



Name	Instance ID	Instance state	Instance type	Status check	Alarm status
------	-------------	----------------	---------------	--------------	--------------

⚠ You are not authorized to perform this operation. User: arn:aws:iam::053020126602:user/intern-user1 is not authorized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInstances action

Retry

Diagnose with Amazon Q

Select an instance

Permissions defined in this policy [Info](#)

Copy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

```
1 {  
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8         "ec2:StartInstances",  
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10      ],  
11      "Resource": "arn:aws:ec2:ap-south-1:053020126602:instance/i-0c0d9260e000ebc44"  
12    },  
13     {  
14       "Sid": "VisualEditor1",  
15       "Effect": "Allow",  
16       "Action": "ec2:DescribeInstances",  
17       "Resource": "*"   
18     }  
19  ]  
20 }
```

Able to start/stop chatapp-application

Instances (1/4) Info Last updated 1 minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states < 1 > [Settings](#)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	chatapp-web	i-0a05131750a99b05e	Stopped	t3.micro	You are not authorized to perform this operation. User: arn:aws:iam::053020126602:user/intern-user1 is not authorized to perform: ec2:DescribeVolumes because no identity-based policy allows the ec2:DescribeVolumes action	An unexpected error has occurred
☐	chatpp-bastion	i-02d04cda008b4f747	Stopped	t3.micro	You are not authorized to perform this operation. User: arn:aws:iam::053020126602:user/intern-user1 is not authorized to perform: ec2:DescribeVolumes because no identity-based policy allows the ec2:DescribeVolumes action	An unexpected error has occurred
☐	chatapp-datab...	i-0ad518b7e11a45b0c	Stopped	t3.micro	You are not authorized to perform this operation. User: arn:aws:iam::053020126602:user/intern-user1 is not authorized to perform: ec2:DescribeVolumes because no identity-based policy allows the ec2:DescribeVolumes action	An unexpected error has occurred
☒	chatapp-appli...	i-0c0d9260e000ebc44	Running	t3.micro	You are not authorized to perform this operation. User: arn:aws:iam::053020126602:user/intern-user1 is not authorized to perform: ec2:DescribeVolumes because no identity-based policy allows the ec2:DescribeVolumes action	An unexpected error has occurred

Rest of the EC2 instances are not editable-start/stop.

Failed to describe volume vol-02bffa14bb4df88c8 [Diagnose with Amazon Q](#) [Close](#)
You are not authorized to perform this operation. User: arn:aws:iam::053020126602:user/intern-user1 is not authorized to perform: ec2:DescribeVolumes because no identity-based policy allows the ec2:DescribeVolumes action

Notifications 2 0 0 0 0 [Settings](#)

Instances (1/4) Info Last updated 2 minutes ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states < 1 > [Settings](#)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	chatapp-web	i-0a05131750a99b05e	Stopped	t3.micro	You are not authorized to perform this operation. User: arn:aws:iam::053020126602:user/intern-user1 is not authorized to perform: ec2:DescribeVolumes because no identity-based policy allows the ec2:DescribeVolumes action	An unexpected error has occurred

S3 bucket:

[General purpose buckets](#) [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (0) Info [Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name < 1 > [Settings](#)

Name	AWS Region	Creation date
------	------------	---------------

You don't have permissions to list buckets [Diagnose with Amazon Q](#)
After you or your AWS administrator has updated your permissions to allow the s3:ListAllMyBuckets action, refresh this page. Learn more about [Identity and access management in Amazon S3](#)

Install the latest PowerShell for new features and improvements! <https://aka.ms>

```
PS C:\Users\260016558> aws configure
AWS Access Key ID [None]: AKIAQYWB65WFMIMDXA3D
AWS Secret Access Key [None]: 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
Default region name [None]: ap-south-1
Default output format [None]: json
PS C:\Users\260016558>
```

```
PS C:\Users\260016558> aws s3 ls
An error occurred (AccessDenied) when calling the ListBuckets operation: User: arn:aws:iam::053020126602:user/intern-user1 is not authorized to perform: s3:ListAllMyBuckets
because no identity-based policy allows the s3:ListAllMyBuckets action
PS C:\Users\260016558> echo "Hello S3!" > file1.txt
PS C:\Users\260016558> aws s3 cp file1.txt s3://iam-s3-intern1/
upload failed: .\file1.txt to s3://iam-s3-intern1/file1.txt An error occurred (AccessDenied) when calling the PutObject operation: User: arn:aws:iam::053020126602:user/inte
rn-user1 is not authorized to perform: s3:PutObject on resource: "arn:aws:s3:::iam-s3-intern1/file1.txt" because no identity-based policy allows the s3:PutObject action
PS C:\Users\260016558>
```

ACTIVITY 2 – IAM Groups & Permission Inheritance

Objective

Understand why groups exist and how permissions flow.

Tasks

1. Create two IAM groups:

BackendTeam

DatabaseTeam

2. Add intern-user1 to BackendTeam

3. Attach AWS managed policy:

○ AmazonS3ReadOnlyAccess to BackendTeam

4. Login as intern-user1 and:

○ List S3 buckets (should work)

○ Upload file (should fail)

Adding 2 groups and adding user into BackendTeam with AmazonS3ReadOnlyAccess

BackendTeam user group created.

View group

✕

Summary

Edit

User group name

BackendTeam

Creation time

January 28, 2026, 00:32 (UTC+05:30)

ARN

arn:aws:iam::053020126602:group/BackendTeam

Users (1)

Permissions

Access Advisor

Users in this group (1)

Remove

Add users

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

< 1 >

⚙

☐

User name

▲

Groups

Last activity

▼

Creation time

▼

☐

[intern-user1](#)

1

11 minutes ago

1 hour ago

BackendTeam user group created.

View group

✕

User group name

BackendTeam

Creation time

January 28, 2026, 00:32 (UTC+05:30)

ARN

arn:aws:iam::053020126602:group/BackendTeam

Users (1)

Permissions

Access Advisor

Permissions policies (1)

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type

All types

< 1 >

⚙

☐

Policy name

▲

Type

▼

Attached entities

▼

☐

AmazonS3ReadOnlyAccess

AWS managed

1

Viewing the S3 bucket from intern-user1, but uploading file is disabled

intern-user1

ⓘ

⚙

🔍

General purpose buckets

All AWS Regions

Directory buckets

General purpose buckets (1)

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

⚙

☐

Name

▲

AWS Region

▼

Creation date

▼

☐

[iam-s3-intern1](#)

Asia Pacific (Mumbai) ap-south-1

January 27, 2026, 23:16:40 (UTC+05:30)

```
PS C:\Users\260016558> aws s3 ls
2026-01-27 23:16:41 iam-s3-intern1/
PS C:\Users\260016558> aws s3 cp file1.txt s3://iam-s3-intern1/
upload failed: .\file1.txt to s3://iam-s3-intern1/file1.txt An error occurred (AccessDenied) when calling the PutObject operation:
arn-user1 is not authorized to perform: s3:PutObject on resource: "arn:aws:s3:::iam-s3-intern1/file1.txt" because no identity-based
PS C:\Users\260016558>
```

Questions to Answer

● Why upload is denied?

Because the IAM group (**BackendTeam**) only has **read-only access** to S3, not write access. The attached policy is: AmazonS3ReadOnlyAccess. This policy allows only: s3:ListBucket, s3:GetObject, s3:GetBucketLocation. It does NOT include: s3:PutObject any write/delete actions

● Where did the permission come from?

The ability to list S3 buckets came from: The IAM Group “BackendTeam” because it has the attached policy: AmazonS3ReadOnlyAccess (AWS Managed Policy) for IAM user intern-user1

ACTIVITY 3 – Custom Policy Using Visual Editor

Objective

Create a least-privilege custom policy

Scenario

Backend team needs:

- Read & write objects in one specific S3 bucket
- No delete permission

Tasks

1. Create a policy using Visual Editor

2. Service: S3

3. Allow:

- ☐ GetObject
- ☐ PutObject
- ☐ ListBucket

4. Restrict access to:

my-backend-bucket

5. Attach policy to BackendTeam

6. Test:

○ Upload file → allowed

○ Delete file → denied

Expected Outcome

● Fine-grained access

● No admin permissions

In resource:

* means:

- Only objects inside this bucket
- No other bucket allowed

deepika-backend-bucket-policy

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Sid": "VisualEditor0",  
6       "Effect": "Allow",  
7       "Action": [  
8         "s3:PutObject",  
9         "s3:GetObject",  
10        "s3:ListBucket"  
11      ],  
12      "Resource": [  
13        "arn:aws:s3:::deepika-bucket-backend",  
14        "arn:aws:s3:::deepika-bucket-backend/*"  
15      ]  
16    }  
17  ]  
18 }
```

```
PS C:\Users\260016558> aws s3 cp file1.txt s3://deepika-bucket-backend/  
upload: .\file1.txt to s3://deepika-bucket-backend/file1.txt  
PS C:\Users\260016558> aws s3 rm s3://my-backend-bucket/file1.txt  
delete failed: s3://my-backend-bucket/file1.txt An error occurred (AccessDenied) when calling the DeleteObject operation: Access Denied  
PS C:\Users\260016558>
```

deepika-bucket-backend
[Info](#)

[Objects](#)
[Metadata](#)
[Properties](#)
[Permissions](#)
[Metrics](#)
[Management](#)
[Access Points](#)

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

<
1
>

☐	Name	Type	Last modified	Size	Storage class
☐	file1.txt	txt	January 28, 2026, 01:09:41 (UTC+05:30)	24.0 B	Standard

ACTIVITY 4 – Custom Policy Using JSON Editor

Objective

Understand IAM policy JSON structure.

Scenario

Frontend team can:

- Start/Stop only one EC2 instance
- Cannot access other instances

Tasks

1. Create a policy using JSON editor

2. Allow actions:

- ☐ ec2:StartInstances
- ☐ ec2:StopInstances
- ☐ ec2:DescribeInstances

3. Restrict resource to:

One specific EC2 instance ARN

4. Create group:

FrontendTeam

5. Attach policy

6. Test by logging in as frontend user

Expected Outcome

- Only one EC2 visible & controllable

```
deepikaa-instance-frontend-policy
```

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "ec2:StartInstances",
8         "ec2:StopInstances"
9       ],
10      "Resource": "arn:aws:ec2:ap-south-1:053020126602:instance/i-0c0d9260e000ebc44"
11    },
12    {
13      "Effect": "Allow",
14      "Action": "ec2:DescribeInstances",
15      "Resource": "*"
16    }
17  ]
18 }
```

Able to start/stop chatapp-application

Instances (1/4) Info

Last updated 1 minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	chatapp-web	i-0a05131750a99b05e	Stopped	t3.micro	You are not authc	An unexpecter
☐	chatpp-bastion	i-02d04cda008b4f747	Stopped	t3.micro	You are not authc	An unexpecter
☐	chatapp-datab...	i-0ad518b7e11a45b0c	Stopped	t3.micro	You are not authc	An unexpecter
☒	chatapp-appli...	i-0c0d9260e000ebc44	Running	t3.micro	You are not authc	An unexpecter

Rest of the EC2 instances are not editable-start/stop.

Failed to describe volume vol-02bffa14bb4df88c8

You are not authorized to perform this operation. User: arn:aws:iam::053020126602:user/intern-user1 is not authorized to perform: ec2:DescribeVolumes because no identity-based policy allows the ec2:DescribeVolumes action

Diagnose with Amazon Q

Notifications 2 0 0 0 0

Instances (1/4) Info

Last updated 2 minutes ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	chatapp-web	i-0a05131750a99b05e	Stopped	t3.micro	You are not authc	An unexpecter